Understanding hypertension dynamics in the South African population: a latent variables approach to the analysis and comparison of data from multiple surveys

By

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2.1.3 Population estimates of blood pressures

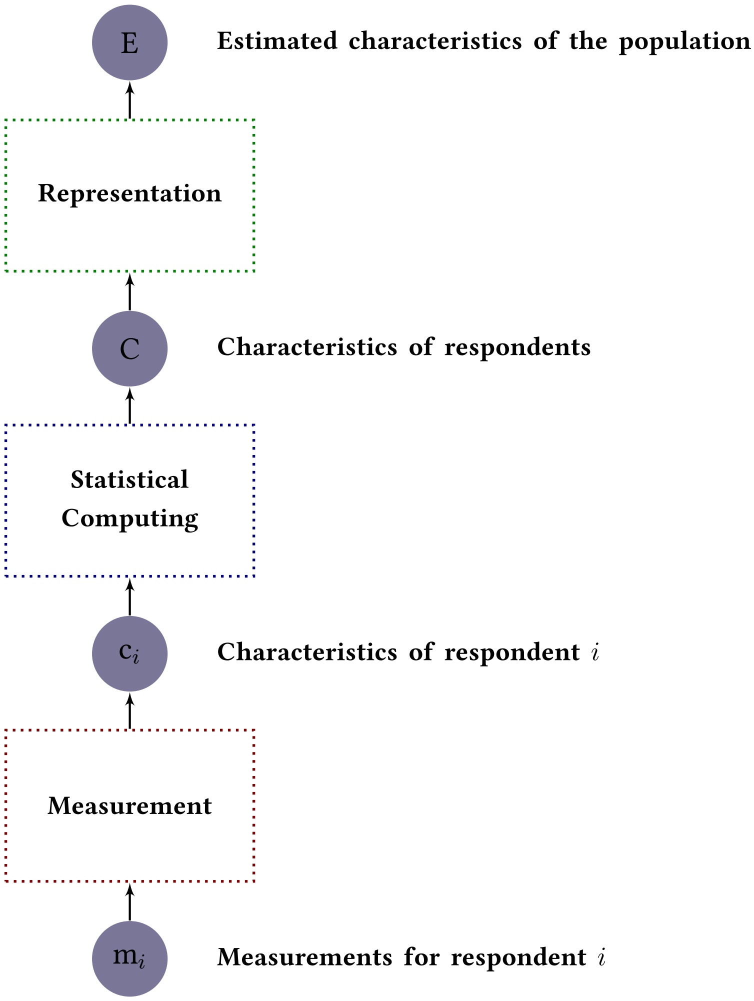
Blood pressure is commonly measured in epidemiological studies of cardiovascular disease (CVD), and the recorded values used for the estimation of population means and marginal distributions, prevalence of the various stages of hypertension, and relationships with other variables of interest.

Survey estimates, even from studies carried out with great attention to quality, are affected by error, and, in the case of surveys involving measurements of blood pressure, the potential magnitude of this error has been of concern for many years.[73–75]

A general representation of the process of survey estimation of blood pressure related quantities[8] according to the total survey error (TSE) framework described by Groves and colleagues is shown in figure 2.6.[76–78]

In the figure, the overall process is represented as the result of two different and conceptually distinct inferential processes[9]. The first process (*measurement*) links the 'imperfect' data on blood pressure recorded for a single participant to a survey to his or her 'true' blood pressure which is not directly observable, while the second inferential process (*representation*) links the true individual values of blood pressure with the population parameters of interest. The two inferential processes are connected by a deterministic procedure (*statistical computing* in the figure), which consists in the calculation of an appropriate summary measure of the values recorded for each individual in the sample (*statistic*).

Figure 2.6: Logical steps in the inferential process of survey estimation.



Adapted from Groves et al. [77]

2.1.3.1 *Measurement*

Figure 2.7 details the logical steps in the measurement process which consists in the estimation of the *unobserved* true blood pressure from a set of one or more *recorded readings*, each of them is an imperfect measure of the patient's casual blood pressure.

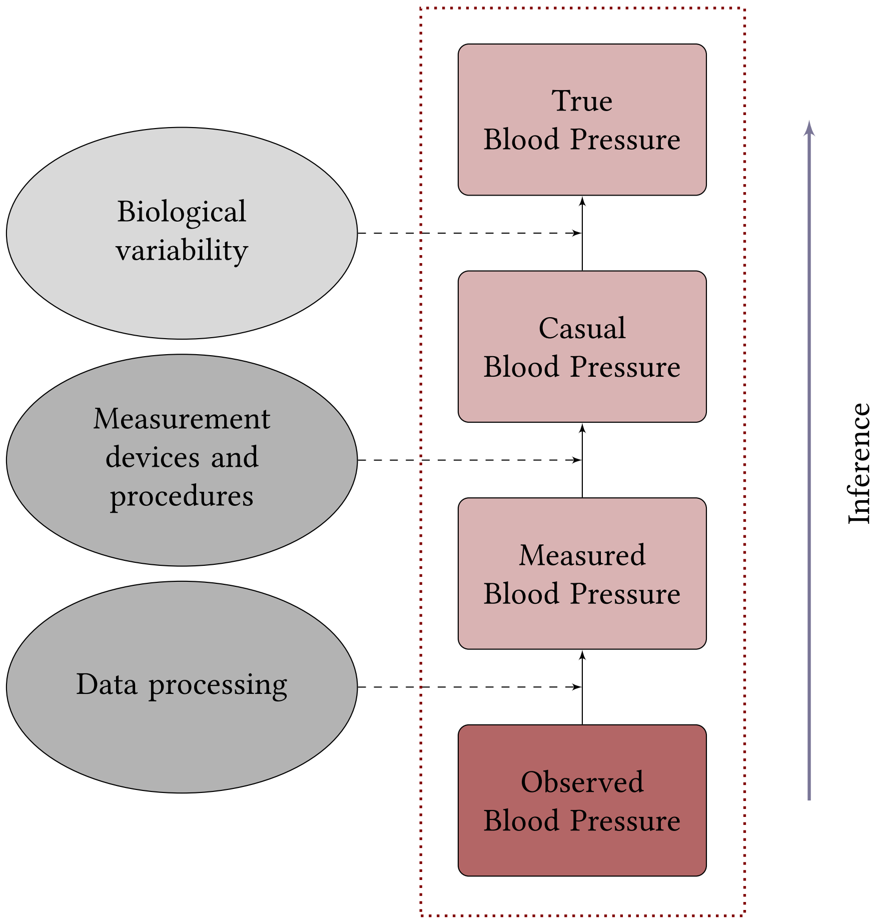
In this conceptualization, we distinguish four different type of blood pressure values:

- BP and *Casual* BP, which are respectively the 'true' individual blood pressure and its value at a specific point in time, as defined in section 2.1.1.
- *Measured* BP, which is the value of BP observed as a result of a measurement procedure.

It is often indicated as a BP *reading*.

- *Recorded* BP, which is the value of BP that is actually recorded.

Figure 2.7: Measurement process and sources of error.



In general, recorded BP differs from observed BP which in turn differs from casual BP and true BP. These discrepancies, collectively indicated as *measurement error* are due to different phenomena.

- The difference between true pressure and casual BP is due to *biological variability*, i.e. to the existence of the 'nuisance component' discussed in section 2.1.1.
- The difference between casual BP and measured BP is due to a multiplicity of reasons which relate to the incorrect conceptualisation and/or application of the *measurement procedures*, and to the use of inaccurate or unrealiable *measurement instruments*;

- The difference between measured and recorded BP is due to the *editing procedures* applied to the actual readings from the measurement instruments. These procedures can be *formal* (such as data cleaning procedures applied to the 'raw' datasets resulting from the fieldwork to exclude or impute implausible values) or *informal* (such as the rounding of the readings from measurement instruments to the nearest even number, because of the well know preference of human observers for numbers that end with some digits rather than others). Both translate into differences between the values recorded and used for subsequent analyses and those actually measured.

Biological variability, errors due to measurement procedures/instruments and processing errors, all belong to the category of *measurement errors*, in the sense that they make the recorded blood pressure different from the true blood pressure which we are interested in quantifying. It is, however, important to highlight that, while differences between recorded, observed and casual blood pressure are due to methodological errors (potentially removable with more accurate procedures and instruments and better training of observers), the difference between casual and true blood pressure represents a real biological difference that we can estimate and sometimes adjust for, but never eliminate.

2.1.3.2 *Representation*

Moving from measurement of individual BP to epidemiological investigation, when the goal is the estimation of BP means or other parameters at population level, other types of errors must be taken into account.

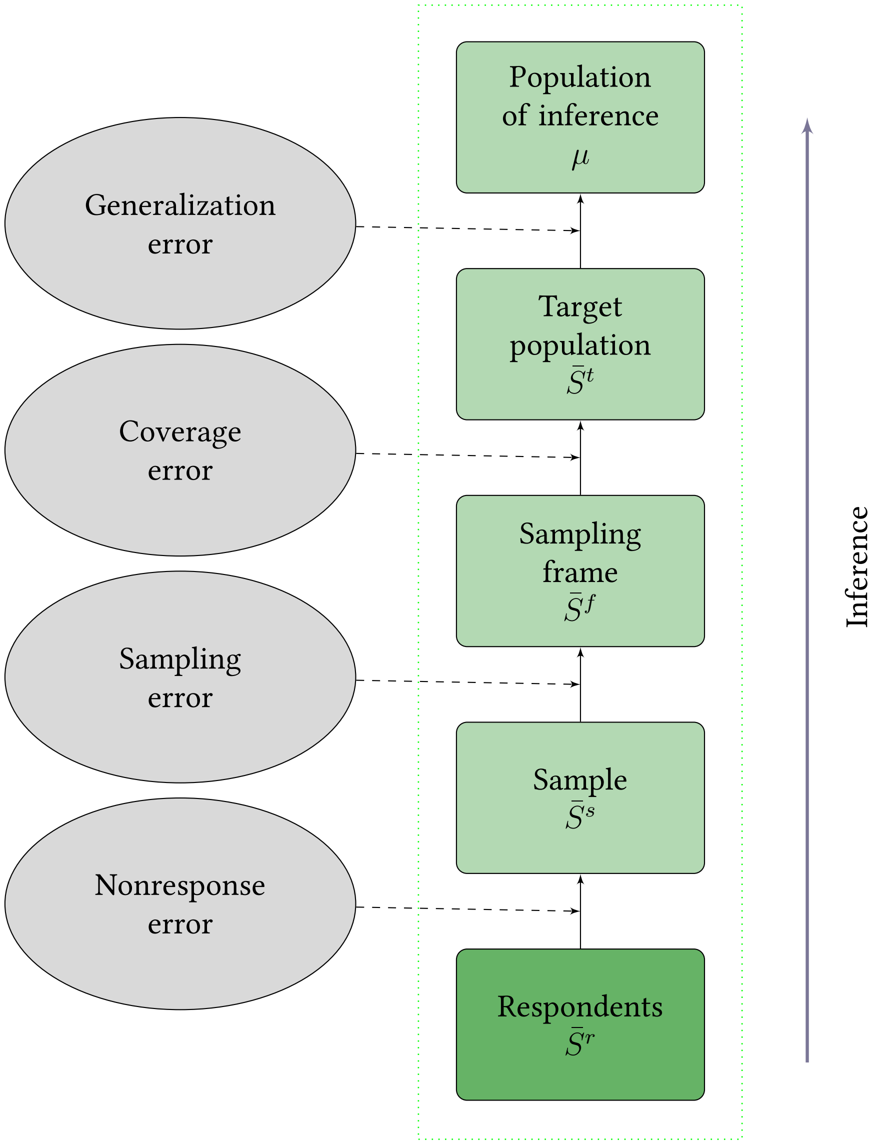
These errors (collectively indicated as *representation error*[76, 77]) are common to all situations where population values of some variables of interest are estimated examining a sample of members of the population, rather than the whole population.

Figure 2.8 shows the logical steps in the representation process that link the value of some statistics summarising the characteristics of a subset of subjects from a larger population (*respondents* or *participants*), to the value of the unobserved population parameter of interest.

In the practical realization of a population survey, the actual respondents are selected from a population of interest (the *population of inference*, to which the researchers aim to generalise their results) though a set of intermediate steps.

The first step entails the definition of the *target population*, i.e. the population that is actually studied. The target and the inferential populations are in general different, and the differences are usually due to logistic/practical reasons. For example, in the South Africa's National Income Dynamics Study (NIDS),[79] the population of inference is the South African population, while the target population formally excludes part of it, and is precisely defined as

Figure 2.8: Representation process and sources of error.



> "*...private households and residents in workers' hostels, convents and monasteries. The frame excludes other collective living quarters such as students' hostels, old age homes, hospitals, prisons and military barracks*" [79, p. 10]

The next step in the process is the identification of the *sampling frame*, that is the totality of members of the target population that have a probability >0 of being selected into the survey sample. In simple cases, the sampling frame consists in a list that includes all subjects in the target population, but in large scale surveys this is generally impossible and other methods are used. For example, in the NIDS, the sampling frame was defined in geographical terms, as a list of small areas subdividing the whole South African territory, and households and individuals were then selected within each area.[79]

From the sampling frame, a *sample* is then selected, which is the subset of individuals from which the measurements of interest (BP in our example) are sought.

Finally, in almost all surveys, the attempt to measure the variables of interest in all the individuals in the sample does not achieve full success. Those for whom the measurements are successfully taken are called *respondents* and are a subset of the whole sample.

The *representation error* — i.e. the total error associated with the estimation of the average BP in the population from a set of BP values measured in the respondents — consists of different elements, conceptualised as the difference of BP estimates within each of the populations and sub-populations defined above. Specifically:

- The difference between the average BP measured among respondents ($\bar{S}^r$) and the average BP among the totality of the sample ($\bar{S}^s$) is indicated as *nonresponse error*. Nonresponse error originates from systematic differences (in the variables of interest) between sampled individuals and respondents.

- The difference between $\bar{S}^s$ and the average BP for all the individuals in the sampling frame ($\bar{S}^f$) is indicated as *sampling error*.

  All statistical estimation procedures are based on the assumption that the sample is randomly selected from the sampling frame, i.e. that all individuals of the sampling frame have a probability >0 and *known* of being selected. Sampling error is the consequence of the ubiquitous violation of this assumption.

- The difference between $\bar{S}^f$ and the average BP for all the individuals in the target population ($\bar{S}^t$) is indicated as *coverage error*.

  The coverage error is the consequence of incomplete or otherwise incorrect sampling frames. For example, if the sampling frame excludes some subjects belonging the target population, these subjects will have a probability 0 of being selected in the sample, which will be no longer representative of the *whole* population.

- The difference between $\bar{S}^t$ and μ is the *generalization error*, i.e. an error that originates from the fact the the target population does not correspond to the population of inference, so that the values calculated in the former cannot be applied to the latter.

2.1.3.3 *Data analysis*

A final source of error in survey estimates (*inferential error*) originates from the estimation procedure itself, i.e. from incorrect assumptions regarding the statistical properties of the variables and/or other modelling assumptions.

For example, error can arise from incorrectly assuming a normal distribution for blood pressure values in the population, while the 'true' distribution is skewed (which is often the case in real populations), or from adopting unsubstantiated assumptions about the relationships between the variables of interest.

2.1.4 Sources of error in population estimates of blood pressure

2.1.4.1 *Biological variability*

2.1.4.1.1 Food intake

Ingestion of food is associated with acute fall in systolic and diastolic blood pressure and increase of heart rate and cardiac output[10].[80–85]

Postprandial reductions in BP are much more common in older than younger people, with a prevalence as high as 36% among those residing in care homes and as 67% in the older hospitalised population.[86]

The magnitude of reduction is modest and of short duration (less than 30 minutes) in younger subjects, but larger on long lasting in older ages.[80, 87–89] A study in a sample of 20 hypertensive patients aged 31 to 66 years found an inverse correlation between drop in blood pressure and age, with a statistically significant correlation coefficient $r = -0.37$.[81] The increase in heart rate, conversely, ha been documented both among young and healthy elder subjects, and magnitudes of 5/6 bpm are common.[84, 85]

The magnitude of the drop in BP observed in different studies varies according to the characteristics of the sample and the experimental conditions, including, non surprisingly, type and quantity of food.[81, 90] A recent systematic review analysed 14 randomised controlled studies on the effect of pharmacological treatment on postprandial change in BP, including 365 subjects aged 41 to 89 years.[86] Among healthy subjects belonging to the control groups, the studies reported postprandial SBP drops between 7 $mmHg$ and 17 $mmHg$, and DBP drops between 5 $mmHg$ and 9 $mmHg$. The observed decreases were as high as 42/14 $mmHg$ (respectively for SBP and DBP) among subjects diagnosed with *autonomic failure*[11]. Studies on younger subjects have usually estimated average postprandial decreases that were smaller (2/3